

FORECASTABLE AND OBSERVABLE REDUCED-ORDER STATES FOR EXTREME VORTEX-GUST AIRFOIL INTERACTIONS

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INTRODUCTION

Extreme vortex-gust encounters generate rapid aerodynamic-load transients through the formation and shedding of leading-edge vortices, a regime relevant to small air vehicles where the gust velocity rivals the flight speed. The response depends on the nonlinear interaction between the incoming vortex, the separated wake, and the phase of the natural shedding cycle [1].

A useful reduced-order state must do two things: retain the vortical structures that govern the transient, and remain predictive when evolved forward, since forecasting depends on the future state rather than its instantaneous reconstruction. Proper orthogonal decomposition (POD) and nonlinear autoencoders (AEs) are designed to reconstruct the field; whether their coordinates also form a dynamically consistent state for gust evolution is less clear.

We compare three reduced-order states: a linear POD basis, an AE [2], and a joint-embedding predictive architecture (JEPA) [3] trained to predict the future reduced state directly, without reconstructing the field. The question is not which reconstructs the flow best, but which provides a compact state whose forward evolution preserves the organisation of the gust

encounter.

METHODOLOGY

The study uses direct numerical simulations at $Re = 5000$ of a NACA 0012 at $\alpha = 14^\circ$ (post-stall), computed with the high-order solver SOD2D. A transient gust is introduced as an idealised inflow perturbation emulating localised unsteady loading: a discrete Taylor vortex of strength G , core diameter D , and wall-normal offset Y convects onto the leading edge and forms a transient leading-edge vortex. The ensemble spans 84 cases, about 378 encounters: training cases, a held-out in-distribution set (42 encounters), and a $|G| = 4$ extrapolation set (24).

To compare methodologies, three encoders map the mid-plane spanwise vorticity field, $\omega_z(x, y, t)$, to a latent $\mathbf{z} \in \mathbb{R}^d$ at two matched budgets, $d = 32$ and $d = 64$: a predictive joint-embedding architecture (JEPA), an AE after [2], and POD. A JEPA does not reconstruct the field; it encodes each frame through a convolutional stem and a small vision transformer over the resulting field patches, $\mathbf{z}_t = E_\theta(\omega_z^t)$, and trains a predictor P_ϕ to advance the latent in time, scoring the prediction in the latent space. With conditioning $\mathbf{c} = (G, D, Y)$ the AE ($d = 64$)

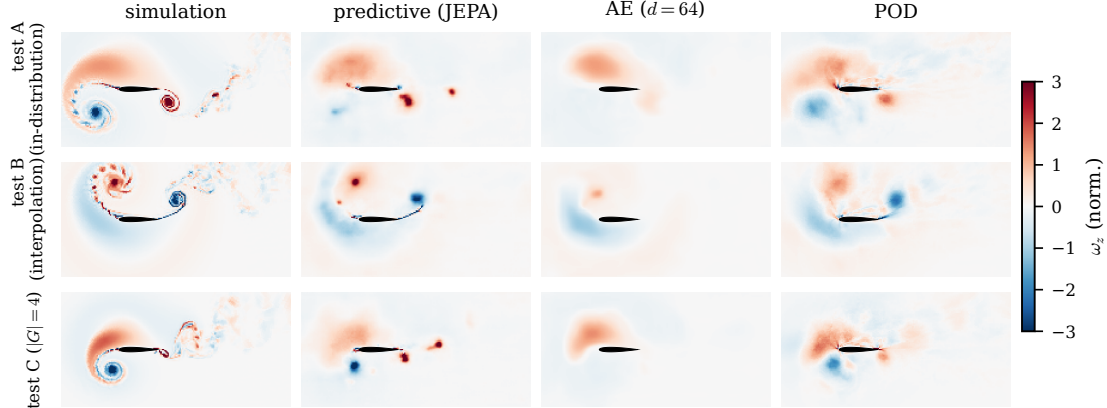


Figure 1: Decoded vorticity at the impact frame. Rows: a test A (in-distribution held-out) encounter at $(G, D, Y) = (+3.0, 1.5, +0.2)$, a test B (interpolation) encounter at $(-3.0, 1.5, -0.1)$, and a test C (extrapolation, $|G| = 4$) encounter at $(+4.0, 1.0, +0.1)$, each the strongest gust in its split. Columns: the simulation, the predictive (JEPA) decode, the matched autoencoder (AE, $d = 64$), and POD.

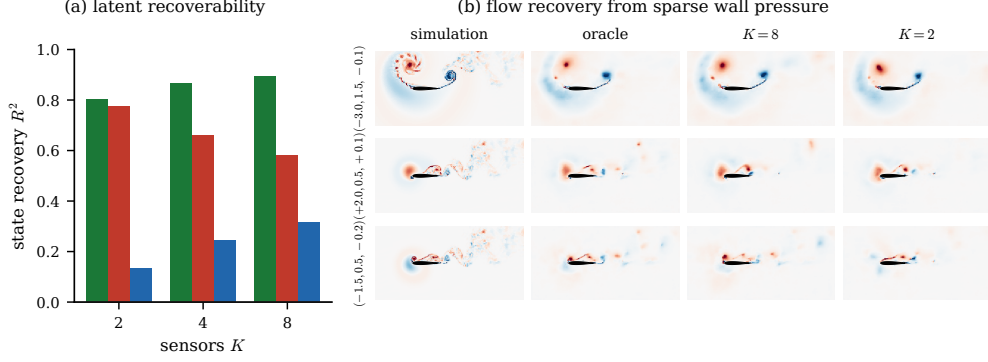


Figure 2: Observability of the reduced state. (a) Held-out latent recovery R^2 from $K \in \{2, 4, 8\}$ TCSI wall-pressure taps at matched $d = 64$ (JEPA green, AE red, POD blue); the predictive state is the most recoverable at every K . (b) Flow recovered on three held-out encounters (rows (G, D, Y)): simulation, oracle decode (decoder on the full-field latent), and the predictive latent estimated from $K = 8$ and $K = 2$ taps; eight taps recover the leading-edge vortex close to the oracle, two the gross impingement.

objective is

$$\mathcal{L} = \sum_t \|P_\phi(\mathbf{z}_{\leq t}, \mathbf{c}) - \mathbf{z}_{t+1}\|^2 + \lambda_R \mathcal{R}(\{\mathbf{z}_t\}) + \lambda_o \mathcal{L}_{\text{obs}}, \quad (1)$$

a latent-prediction error (with a scheduled-sampling rollout term); an anti-collapse regulariser $\mathcal{R}$ (a characteristic-function penalty matching the latent distribution to an isotropic Gaussian, preventing collapse); and a small auxiliary loss $\mathcal{L}_{\text{obs}}$ from two heads reading off the latent the lift C_L and an 80-dimensional wake descriptor (signed patch energies over the wake region plus a 16-bin radial power spectrum), after the observable-augmentation line. No term reconstructs the field. The AE carries the lift head but not the wake head, so the comparison is between families as configured; matched-supervision ablations attribute the wake gain to the predictive objective with wake supervision.

A single conditional transformer predictor, of the same architecture and training fit to each encoder’s own latents, advances the latent conditioned on $\mathbf{c}$ through identity-initialised adaptive layer normalisation; a common probe reads the observables, so that only the encoder differs. We judge each state by *forward physical closure*: rolling that predictor $H = 16$ convective steps from impact and reading six observables off the rolled-out latent, the force coefficients C_L and C_D , the vertical impulse I_y , the wake enstrophy ($\int_{\text{wake}} \omega_z^2 dA$) and the signed circulations $\Gamma^\pm$, compared to the simulation. These combine body-force, integral and distributed-wake information. Reconstruction quality is reported but is not the primary metric.

RESULTS

The representations differ first in what they retain. Figure 1 shows decoded impact-frame vorticity for held-out encounters: the predictive representation preserves the leading-edge-vortex impingement and wake organisation more consistently than the AE, which smooths or collapses the vortical structures, while POD is amplitude-accurate but misplaces the dominant vortices. The $|G| = 4$ case is hardest for all three, consistent with the mid-plane representation near a more three-dimensional regime.

The more important distinction appears under forward evolution. With the common predictor, the predictive representation gives the most consistent closure of the wake observables, especially wake enstrophy and signed circulation. This is not merely better instantaneous reconstruction; it is geometric. Under rollout the AE latent progressively leaves the region of

latent space occupied by the training data, whereas the predictive state stays near its learned manifold, and this reduced drift preserves wake information over the horizon.

Finally, the states can be estimated online from sparse wall pressure. A target-conditioned procedure (TCSI) selects a few surface taps and uses their pre-impact histories to estimate the latent. The predictive state is the most recoverable of the three (Figure 2a); from eight taps it retains the leading-edge vortex and shear layer close to the full-field oracle, and two taps capture the gross impingement (Figure 2b).

CONCLUSIONS

Reconstruction accuracy alone is not sufficient: a state that reconstructs the instantaneous field may still fail when propagated forward if its latent geometry is incompatible with the flow dynamics. Representations trained around future evolution provide a more forecastable wake state, supported by several diagnostics: reduced rollout drift, a simpler cyclic organisation of the encounter, agreement with optimal-transport measures of vortical motion, and better recovery from wall pressure. Forecastability and observability appear to be closely related properties of a physically meaningful reduced-order state.

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